

# INTERNSHIP PROJECT REPORT

## Task 4: Real-Time AI Vision & Object Tracking System

**Intern Name:** Ayesha Aftab

**Internship Domain:** Artificial Intelligence

**Organization:** CodeAlpha

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### 1. Project Overview

The goal of this project was to build a sophisticated Computer Vision dashboard capable of real-time object detection and persistent tracking. Completed as **Task 4** of the **CodeAlpha AI Internship**, this system—titled "Vision AI"—utilizes state-of-the-art Deep Learning models to identify and monitor multiple objects simultaneously through live webcam feeds or static images.

### 2. Technical Specifications

- **AI Engine:** YOLOv8 (You Only Look Once) by Ultralytics.
- **Core Model:** yolov8n.pt (Nano model for high-speed real-time inference).
- **Computer Vision:** OpenCV (cv2) for frame capture and image processing.
- **User Interface:** customtkinter for a modern, hardware-accelerated "Dark Tech" dashboard.
- **Backend Integration:** PIL (Pillow) for seamless AI-to-GUI image conversion.

### 3. Key Features

- **Live Stream Analytics:** Real-time object tracking from webcam input with persistent ID assignment.
- **High-Precision Detection:** Identifies diverse classes (humans, vehicles, electronics, etc.) with confidence scoring.
- **Intel Analysis Panel:** A dedicated terminal-style output log that displays live metadata for every detected object.
- **Multi-Mode Input:** Supports both continuous live video streams and single-image forensic analysis.

- **Professional UI Design:** A custom "Control Center" layout featuring an elegant sidebar, central vision screen, and emerald-green intelligence logs.

#### 4. Methodology

The system was developed using a modular architecture:

1. **Model Integration:** Integrated the YOLOv8 tracking pipeline to handle both detection and object persistence across consecutive frames.
2. **Vision Loop:** Implemented a recursive processing loop (`process_loop`) that captures frames, runs inference, and updates the UI at high FPS (Frames Per Second).
3. **UI/UX Design:** Leveraged `customtkinter` to create a grid-based responsive layout, using hex-coded deep black and tech-blue themes for a professional industry look.
4. **Metadata Extraction:** Developed a parsing logic to extract class names and confidence levels from the AI results and display them in a scrollable tracking box.
5. **Session Management:** Built safe start/stop mechanisms to release hardware resources (webcam) when the application is shut down.

#### 5. Results & Conclusion

The **Vision AI Control Center** demonstrates the power of modern real-time inference engines. The system maintains a high detection accuracy ( $\text{conf} > 0.4$ ) while providing a user-friendly interface for monitoring. This project successfully showcases the application of AI in security, automation, and real-time data analysis, meeting all professional criteria for the CodeAlpha internship.